

10 Utility Functions Quiz ANS

1. What does a Bernoulli utility function represent?

- A) Preferences over outcomes using numerical values
- B) Only objective probabilities
- C) The frequency of failures
- D) A legal duty

ANS: A) Preferences over outcomes using numerical values

2. What is expected value?

- A) The utility assigned to the worst outcome
- B) The probability-weighted average of monetary or numerical outcomes
- C) The most likely outcome only
- D) A person's reported preference

ANS: B) The probability-weighted average of monetary or numerical outcomes

3. What is expected utility?

- A) The sum of unweighted outcomes
- B) The utility of the average outcome
- C) The probability-weighted average of utilities of possible outcomes
- D) The maximum payoff regardless of probability

ANS: C) The probability-weighted average of utilities of possible outcomes

4. What do von Neumann-Morgenstern utility functions represent?

- A) Preferences over certain outcomes only
- B) A model's training loss
- C) A social-welfare ranking only
- D) Preferences over lotteries satisfying specified axioms

ANS: D) Preferences over lotteries satisfying specified axioms

5. What is a lottery in decision theory?

- A) A probability distribution over possible outcomes
- B) A guaranteed monetary gain
- C) An outcome with no uncertainty
- D) A voting system

ANS: A) A probability distribution over possible outcomes

6. What does the St. Petersburg paradox challenge?

- A) The definition of probability
- B) The idea that people simply maximize expected monetary value
- C) The existence of risk
- D) The use of logarithms

ANS: B) The idea that people simply maximize expected monetary value

7. Why can a logarithmic utility function help explain the St. Petersburg paradox?

- A) It makes every outcome equally likely
- B) It creates infinite wealth
- C) Diminishing marginal utility keeps expected utility finite
- D) It removes uncertainty

ANS: C) Diminishing marginal utility keeps expected utility finite

8. What is corrigibility in an AI system?

- A) The tendency to maximize one fixed reward
- B) The ability to predict lotteries
- C) The refusal to accept changes
- D) Willingness to allow correction, modification, or shutdown

ANS: D) Willingness to allow correction, modification, or shutdown

9. Why might a simple utility-maximizing coffee-fetching AI resist correction?

- A)** Correction could lower the utility it expects under its current objective
- B)** It has no objective
- C)** It cannot compare outcomes
- D)** Correction always increases its reward

ANS: A) Correction could lower the utility it expects under its current objective

10. What characterizes risk aversion?

- A)** Preferring every gamble
- B)** Preferring a certain outcome to a gamble with the same expected value
- C)** Maximizing expected monetary value exactly
- D)** Ignoring utility curvature

ANS: B) Preferring a certain outcome to a gamble with the same expected value

11. What characterizes risk neutrality?

- A)** Always rejecting uncertainty
- B)** Always choosing the safest outcome
- C)** Evaluating gambles by expected value without additional preference for or against risk
- D)** Preferring lower expected value gambles

ANS: C) Evaluating gambles by expected value without additional preference for or against risk

12. What characterizes risk-seeking behavior?

- A)** Always choosing certainty
- B)** Using linear utility
- C)** Rejecting all positive variance
- D)** Preferring a gamble to a certain outcome with the same expected value

ANS: D) Preferring a gamble to a certain outcome with the same expected value

13. What shape is commonly associated with a risk-averse utility function over wealth?

- A)** Concave
- B)** Linear
- C)** Constant only
- D)** Convex

ANS: A) Concave

14. What does the independence axiom require?

- A)** Preferences must ignore probability
- B)** Mixing two options with the same third lottery should preserve their ranking
- C)** Every lottery must be equally preferred
- D)** Utilities must equal dollars

ANS: B) Mixing two options with the same third lottery should preserve their ranking

15. What does the Allais paradox illustrate?

- A)** People always satisfy expected-utility axioms
- B)** Expected value is undefined
- C)** Observed choices can violate the independence axiom
- D)** Risk attitudes never depend on framing

ANS: C) Observed choices can violate the independence axiom

16. What does the Asian disease framing example demonstrate?

- A)** Equivalent descriptions always produce identical choices
- B)** Probabilities are irrelevant
- C)** People maximize total wealth
- D)** Framing outcomes as gains or losses can reverse preferences

ANS: D) Framing outcomes as gains or losses can reverse preferences

17. Is prospect theory primarily normative or descriptive?

- A)** Descriptive: it models how people tend to make risky choices
- B)** Normative: it defines what everyone ought to choose
- C)** Neither: it excludes behavior
- D)** Legal: it specifies enforceable duties

ANS: A) Descriptive: it models how people tend to make risky choices

18. What are the two broad stages in prospect theory?

- A)** Training and deployment
- B)** Editing and evaluation
- C)** Voting and auditing
- D)** Prediction and control

ANS: B) Editing and evaluation

19. What does the prospect-theory value function evaluate outcomes relative to?

- A)** Absolute final wealth only
- B)** The largest possible gain
- C)** A reference point, with different treatment of gains and losses
- D)** The social average

ANS: C) A reference point, with different treatment of gains and losses

20. What does the prospect-theory weighting function capture?

- A)** Perfect use of objective probabilities
- B)** Only utility of money
- C)** A ban on low-probability events
- D)** Systematic distortion of decision weights relative to objective probabilities

ANS: D) Systematic distortion of decision weights relative to objective probabilities